

SAFETY REQUIREMENTS

1. Purpose and Scope

This document sets out the safety requirements for all robotics activities. It applies to everyone involved: participants, coaches, judges, volunteers, event hosts and organizers, and spectators. Every person shares responsibility for safety. If you see an unsafe condition or behaviour, report it as soon as possible. No score, deadline, or match is more important than everyone's safety.

2. General Rules - For Everyone

- Closed-toe shoes are recommended at all times. Flip-flops, sandals, and open-heel shoes are not recommended.
- Tie back long hair and secure loose clothing, lanyards, and jewelry near moving mechanisms.
- No running in competition areas.
- Follow all instructions from event staff and posted signage.
- Be cautious with food and drinks around robots, playing fields, computers, and other electronics - make sure to clean up after yourself!
- Keep aisles and emergency exits clear at all times - store bags under tables.
- Report any injury, hazard, or damaged equipment to staff immediately.

3. Team Safety Requirements

3.1 Batteries

Damaged or improperly handled batteries cause serious hazards, including fire and potential injuries.

Selecting:

- Zone01 recommends using standard, manufacturer-original or certified battery packs (e.g., official LEGO rechargeable packs). Never use unbranded or suspiciously cheap batteries from unknown online sources.
- Teams with the proper knowledge and safety measures may choose to build their own battery pack. *

Handling:

- Never crush, bend, drop, puncture, or disassemble a battery.
- A battery that is swollen, cracked, leaking, unusually hot, or was recently dropped is unusable. Take it out of service immediately and give it to a coach or organizer.
- Never carry a battery by its cable or leads.
- Keep battery terminals away from metal objects (tools, keys, spiral notebooks) that could short-circuit them.

Soldering:

- Never solder directly on battery cells or packs.
- Any soldering on other components must be done by or under the direct supervision of an adult, in a ventilated area, on a fire-resistant surface, with eye protection, away from batteries and flammable materials.
- Soldering irons stay in their holders when not in hand and remain hot after being unplugged.

Charging:

- Charge only with the charger designed for that battery.
- Charge in a supervised, ventilated area on a non-flammable surface - never in a closed bag, under papers, or overnight unattended.
- Disconnect batteries at the end of each session.

Storing and transporting:

- Store and transport batteries in a rigid container with terminals protected, away from heat and direct sun.
- Do not leave batteries in a hot vehicle.

** Custom Batteries:*

- Custom packs must include protection circuitry - at minimum a toggle switch and a fuse - and other measures to prevent any short circuit, such as insulated terminals, secure cell mounting, and protected wiring.
- A custom pack is the team's responsibility: it must be assembled and maintained under the supervision of an adult with battery-handling knowledge
- Organizers may inspect it and refuse any pack that appears unsafe.
- You MUST specify whether you will be using a custom battery in the team registration form ... TBD

3.2 Electricity

- Inspect cords and chargers before use - do not use anything with cracked insulation, exposed wire, or a damaged plug.
- Never plug an extension cord or power bar into another extension cord or power bar.
- Do not overload outlets or power bars; respect their rated capacity.
- Use grounded (three-prong) connections as designed.

- When travelling to competitions abroad, check voltage before plugging in: North America uses 120 V, most of Europe and Asia uses 230 V. Verify that your chargers and power bars are rated for the local voltage ("100–240 V" on the label) and use proper plug adapters. A 120 V-only device plugged into 230 V can overheat or catch fire.

3.3 Moving Parts, Sharp Pieces, and Tools

- Some challenges allow for non-LEGO elements and may include metal parts or soldered components. Assume metal pieces may have sharp edges or points. Handle these elements as described in the challenge rules or as directed by staff, and do not modify, disassemble, or attempt to repair them.
- Report problems to staff as soon as they are noticed.
- Keep fingers, hair, and clothing away from gears, wheels, and fast-rotating parts.
- Carry robots with both hands, supporting them from below, with mechanisms secured so nothing can swing or fall.
- Electric and hand tools may be used only with coach or staff supervision. Use the right tool for the job, check its condition first, cut away from your body, and wear eye protection when there is any risk of flying debris.
- Never throw parts, tools, or equipment.
- Keep guards and protective covers in place on any equipment that has them.

3.4 Liquids

- No liquids on tables holding robots, electronics, computers, or playing fields. Water bottles stay closed, on the floor or a separate table.
- If liquid spills on or near electronics: do not touch the equipment, disconnect power at the outlet if it is safe to do so, and notify staff immediately.

3.5 Health and Conduct

- Drink enough water during the day and know where the water stations are located.

- Competition days are long: eat properly, and sleep well the night before.
- Pace yourself. If you feel unwell, dizzy, or overwhelmed, tell your coach - stepping away is always allowed. Look out for your teammates too.
- No running, climbing, or standing on chairs and tables.
- Lift heavy material (playing fields, bins) with a partner, using your legs and not your back - use carts when available.

4. Coaches

Safety supervision is part of the coach's role, and much of it happens before the event.

Before the event:

- Review this document with your team.
- Ensure every team member has submitted a signed consent and release form.
- Inspect your team's batteries, chargers, cords, and tools; leave damaged equipment at home.
- Verify your robot has no exposed sharp edges or pinch points.
- Pack spares: charged batteries, the correct chargers, and plug adapters if travelling.
- Confirm emergency contact information for each team member is up to date and accessible to you at the event.
- If your team is using a custom battery, you are responsible for ensuring proper construction and enforcing strict safety measures.

At the event (if you accompany your team):

- Supervise your team at all times; you are the first point of contact for any incident involving your team.
- Enforce the battery, electricity, and liquid rules at your team's table.
- Know the location of first aid, emergency exits, the water station, and the event coordinator.
- Model the behaviour you expect from your team.

5. Judges

Detecting, reporting, and preventing safety risks on the competition floor is part of the judging role.

Watch for; on robots and playing fields:

- Unprotected or exposed sharp edges and points.
- Loose parts that could detach during a match.
- Exposed or damaged wiring; swollen or damaged batteries.
- Overheating: hot smells, discoloured components, or any sign of smoke - interrupt the match immediately if detected.
- Unsafe behaviour around the field: reaching into a running robot's path, crowding, running.

If you detect a risk:

- Stop the match or activity if there is any immediate danger. A match can always be replayed, but an injury cannot be undone.
- Report the issue to the event coordinator. Minor issues (a sharp edge, a loose part) go to the team's coach for correction before the robot runs again.

6. Hosts and Organizers

Batteries and fire:

- Know the venue's fire procedures and the location of extinguishers before the event opens.
- If a battery overheats, smokes, or catches fire: evacuate the immediate area and alert everyone nearby.

- A trained adult may attempt to extinguish only a very small fire with an appropriate extinguisher; otherwise evacuate and call 911. Lithium fires produce toxic smoke - when in doubt, get out.
- Do not move a smoking battery with bare hands.
- If possible, use a fire blanket to contain and put out a small fire. Do this **ONLY** AFTER turning off the power source. Only turn off the power source if it is safe to do so.
- Designate a safe outdoor location where a damaged (non-burning) battery can be isolated in a non-flammable container.
- Take note of teams that are using custom batteries. Make sure their locations are known by the organizers in case of emergency.

Electricity:

- Plan power distribution before the event: enough circuits and outlets for team tables so that teams are not tempted to daisy-chain.
- Install extension cords along walls where possible - tape down any cords crossing a walkway.
- Verify cords and power bars are rated for the load and in good condition.

Health and venue:

- Ensure a stocked First Aid Kit is accessible and its location is announced and posted
- Identify those who are First-Aid Certified on site.
- Ensure access to a water fountain.
- Keep emergency exits clear and verify the assembly point before doors open.
- Provide carts or platforms for moving playing fields and heavy material.
- Provide enough chairs for participants - people sitting on floors and tables creates hazards.

7. Incident and Emergency Protocols

- **Injury or illness:** Stop the activity and notify the nearest staff member. Staff provide or call for first aid, contact the emergency contact on file if needed, and complete an incident report before the end of the day.
- **Equipment malfunction:** Power off if safe to do so, clear the immediate area, and notify staff. The equipment or station returns to service only after inspection.
- **Battery incident:** Follow Section 6 - isolate or evacuate, never improvise.
- **Evacuation:** Leave equipment behind. Coaches gather their team, proceed to the posted assembly point, and report attendance to the event coordinator.